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Changing Molecular Epidemiology and Resistance Patterns of Carbapenem-Resistant *Klebsiella pneumoniae* in California Long-term Acute Care Hospitals

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Background. The epidemiology of carbapenem-resistant *Klebsiella pneumoniae* (CRKP) must be monitored over time to understand resistance dynamics to existing and novel antibiotics.

Methods. Antibiotic susceptibility testing and whole-genome sequencing were performed on clinical CRKP isolates collected from the same California long-term acute care hospital network across 2 time periods: 2014–2015 (period 1) and 2021–2023 (period 2). Differences in sequence type (ST) diversity, carbapenem resistance-associated genotypes, and resistance to clinically important antibiotics were evaluated.

Results. Eight hundred eighty-one isolates were collected (period 1, 374; period 2, 507). Study period 1's population was dominated by ST258 (94.1%). Sequence type diversity markedly increased in period 2, with a reduced prevalence of ST258 (41.6%) and an increased abundance of other epidemic lineages: ST14 (27.4%), ST307 (6.9%), and ST17 (6.5%). Changes to circulating resistance genotypes and plasmids reflected the increased ST diversity. Increases in bla_{KPC-3}-harboring isolates (66.0% vs 76.1%) and noncarbapenemase-producing isolates (2.7% vs 12.6%) aligned with decreases in bla_{KPC-2}-harboring isolates (30.5% vs 6.3%). Statistically significant differences in resistance patterns were also observed between study periods, including colistin (35.0% vs 20.2%), fosfomycin (27.0% vs 13.4%), meropenem–vaborbactam (9.4% vs 2.0%), and ceftazidime–avibactam (1.1% vs 9.1%). Differential patterns of resistance emergence and spread across circulating STs largely drove resistance patterns in study period 2.

Conclusions. The comparison of clinical CRKP isolates collected from the same healthcare network and separated by nearly a decade revealed increasing ST diversity, mainly due to the introduction of numerous epidemic lineages, which coincided with differing patterns of circulating resistance-associated genotypes and antibiotic resistance.

Keywords. carbapenem-resistant *Klebsiella pneumoniae*; antibiotic resistance; whole-genome sequencing; epidemiology; long-term acute care hospital.

The emergence and worldwide dissemination of multidrug-resistant organisms, such as carbapenem-resistant *Klebsiella pneumoniae* (CRKP), is a significant public health threat [1]. Infections caused by CRKP represent a major challenge due to their clinical complexity and high mortality rates [2]. These challenges are heightened in high-acuity populations with extensive healthcare exposure and frequent antibiotic utilization, such as residents of long-term acute care hospitals (LTACHs) [3–5].

The ability to treat infections caused by carbapenem-resistant Enterobacterales (CRE), such as CRKP, has been enhanced by the introduction of several antibiotics, including novel β-lactam/β-lactamase inhibitor (BL/BLI) combinations and cefiderocol [6–8]. The availability of such agents has been a major contributor to global changes in antibiotic consumption patterns and the treatment of CRE infections over the past decade [9–11].

Genomic investigations have shown that the regional epidemiology of CRE can change due to strain importation, clonal expansion, and selective pressures [12–14]. Prior research has

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indicated that genetic backgrounds can have differential propensities for resistance to emerge and spread, suggesting that certain lineages may be more primed to respond to shifting antibiotic formularies [15–18]. To preserve the long-term efficacy of promising novel antibiotics, investigations into the prevalence and dynamics of resistance, especially in high-risk populations, are of utmost importance.

Here, we compare 2 period prevalence studies, both of which performed comprehensive clinical isolate collection of CRKP across a California LTACH network, separated by nearly a decade and the availability of several novel antibiotics. Whole-genome sequencing and antibiotic susceptibility testing enabled assessment of how the molecular epidemiology and resistance patterns changed within a large geographic catchment over a period of shifts in antibiotic consumption.

METHODS

Study Population

We performed 2 period prevalence studies. As part of a prior study, clinical CRKP isolates were obtained from 1 August 2014, to 25 July 2015 (period 1), at 12 southern California LTACHs [19, 20]. All patients with a clinical CRKP culture were included. To capture the diversity of circulating CRKP while balancing the number of isolates included per patient, patients could contribute multiple isolates, but only their first isolate per calendar quarter was retained for this analysis (Supplementary Figure 1). We subsequently performed a comprehensive clinical isolate collection from the same LTACH network from 7 June 2021, to 31 December 2023 (period 2) using the same specifications. In period 2, 1 LTACH was added to the network.

Microbiological Methods

In both periods, all samples identified as CRKP by the network's central microbiology laboratory, defined as laboratory-confirmed *K. pneumoniae* demonstrating resistance to ertapenem, imipenem, and/or meropenem, were shipped to the study laboratory at the University of Pennsylvania. Confirmatory identification was conducted using VITEK-MS (bioMérieux, Durham, NC).

Antibiotic susceptibility testing was performed on 2014–2015 study isolates using a custom SENSITIRE broth microdilution panel (Thermo Scientific, Waltham, MA), as described elsewhere [15]. For the 2021–2023 study, we used a similar SENSITITRE plate (STYMDRGNXXF, Thermo Scientific). Cefiderocol susceptibility testing was performed by disk diffusion using 30 µg disks on Mueller–Hinton agar (Thermo Scientific).

Antibiotic resistance was defined as a minimum inhibitory concentration in intermediate or resistant categories, expect where otherwise noted. Interpretation criteria are reported in the Supplementary Methods and Supplementary Table 1.

Whole-Genome Sequencing

Study period 1 isolates underwent whole-genome sequencing on an Illumina HiSeq 2500 instrument at the PennCHOP Microbiome Center, as described elsewhere [20]. Study period 2 isolates underwent whole-genome sequencing on an Illumina NovaSeq X instrument at the University of Michigan's Advanced Genomics Core. Sequencing data are available under BioProjects PRJNA415194 and PRJNA1429228.

Raw sequencing data underwent quality control, genome assembly, and taxonomical classification using an in-house pipeline (https://github.com/Snitkin-Lab-Umich/QCD) [21–23]. Species and sequence type (ST) were identified using skani v.0.2.1 and Kleborate v.3.1.3, respectively [24, 25]. Antibiotic resistance genes and mutations in the major porins *ompK35* and *ompK36* were detected using AMRFinderPlus v.4.0.23 [26], while porin truncations were assessed using Kleborate. Carbapenemase-containing plasmids were reconstructed from short-read assemblies using MOB-Suite v.3.1.9 [27]. Quality control data for study isolates are in Supplementary Table 2. Details on the phylogenetic reconstruction and phylogeographical contextualization of the 2021–2023 isolates are reported in the Supplementary Methods [16, 28–31].

Clinical Metadata Collection

Clinical metadata were extracted from the healthcare network's electronic medical records, as described previously [19, 20]. The clinical data ascertainment approach was identical across study periods.

Data Analysis and Visualization

First, we compared the demographics and length of stay for subjects from period 1 versus period 2. Next, to characterize the molecular epidemiology of CRKP across study periods, we compared the distributions of STs, carbapenemases, and carbapenemase-containing plasmids. Finally, we compared the proportion of resistant isolates across study periods for each antibiotic. In these comparisons, continuous variables were compared using the Student's *t*-test or the Wilcoxon rank-sum test. Categorical variables were compared using the χ^2 or Fisher's exact test.

All analyses, unless otherwise noted, were performed using R version 4.6.0 (R Core Team, 2025). Tidyverse packages and ggnewscale v.0.5.2 were used to manipulate and visualize the data [32, 33]. Descriptive statistics were calculated using tableone v.0.13.2 [34]. Heatmaps were curated using ComplexHeatmap v. 2.28.0 [35]. Alluvial plots were created using ggalluvial v.0.12.6 [36]. Phylogenetic analysis and visualizations were performed using ape v.5.8-1, ggtree v.4.2.0, phyloAMR v.0.1, and phytools v.2.5-2 [16, 37–39]. Cowplot v.1.2.0 generated multipanel figures [40]. Manuscript code is available at https://github.com/kylegontjes/INTEGRATE-resistance-ms.

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RESULTS

Surveillance of Carbapenem-Resistant *K. pneumoniae* in California Long-term Acute Care Hospitals

As part of a prior study, 374 clinical CRKP isolates were obtained from 329 patients from 1 August 2014, to 25 July 2015 (period 1) across 12 southern California LTACHs [20]. We subsequently performed a new collection of clinical CRKP isolates in the same LTACH network from 7 June 2021, to 31 December 2023 (period 2) using the same methodology as in period 1. During the second collection, 394 patients contributed 507 CRKP isolates.

Patient Characteristics

Baseline characteristics of patients with clinical metadata did not substantively differ across study periods (Supplementary Table 3). The proportion of male patients was similar across study periods (period 1, 170/329 [51.7%]; period 2, 197/379 [52.0%]; χ^2 test $P = 1.00$). Although not collected in period 1, 140 (36.9%) patients were non-Hispanic White in period 2. The median age at first isolate collection was 73.0 years (interquartile range [IQR], 18.0) in period 1 and 69.0 years (IQR, 16.0) in period 2, a statistically significant but not clinically meaningful difference (Wilcoxon rank-sum test $P < .001$). Patients had shorter lengths of stay before their initial specimen collection in period 1 (median [IQR], 20.0 [38.0] days) than in period 2 (median [IQR], 32.0 [60.0] days; Wilcoxon rank-sum test $P < .001$). Finally, the distribution of isolates across the 12 facilities in both study periods did not differ statistically (χ^2 test $P > .05$).

Anatomic Distribution of Isolates

The anatomical distribution of isolates differed between study periods (Fisher's exact test $P < .001$). The relative contribution of wound (6 [1.6%] vs 77 [15.2%]) and blood (30 [8.0%] vs 65 [12.8%]) isolates increased in study period 2, whereas the contribution of respiratory (217 [58.0%] vs 242 [47.7%]) and urine (121 [32.4%] vs 113 [22.3%]) isolates decreased. Ten additional isolates in period 2 were isolated from other anatomical sites.

Increasing Diversity of Carbapenem-Resistant *K. pneumoniae*

Study period 1 had low ST diversity, with only 10 STs detected (Figure 1*A*). ST258 (352 [94.1%]) dominated, with only 4 other nonsingleton STs being identified: ST15 (11 [2.9%]), single-locus variants of ST15 (2 [0.5%]), ST17 (2 [0.5%]), and ST36 (2 [0.5%]). Conversely, a much more diverse population was identified in the 2021–2023 study. Although 36 STs were identified, 23 were found in only one isolate. Most isolates belonged to epidemic lineages, notably ST258 (211 [41.6%]), ST14 (139 [27.4%]), ST307 (35 [6.9%]), ST17 (33 [6.5%]), ST25 (13 [2.6%]), ST35 (13 [2.6%]), and ST11 (9 [1.8%]).

Only 2 STs were identified across both study periods (Figure 1*B*). While ST258 remained the dominant lineage during the 2021–2023 study period, its prevalence decreased markedly (352 [94.1%] vs 211 [41.6%]; χ^2 test $P < 0.001$). The prevalence of epidemic lineage ST17 increased significantly (2 [0.5%] vs 33 [6.5%]; Fisher's exact test $P < .001$). Phylogeographic analysis of epidemic lineages ST14, ST17, ST258, and ST307 uncovered numerous independent introductions into the region of varying magnitude, alongside the clonal

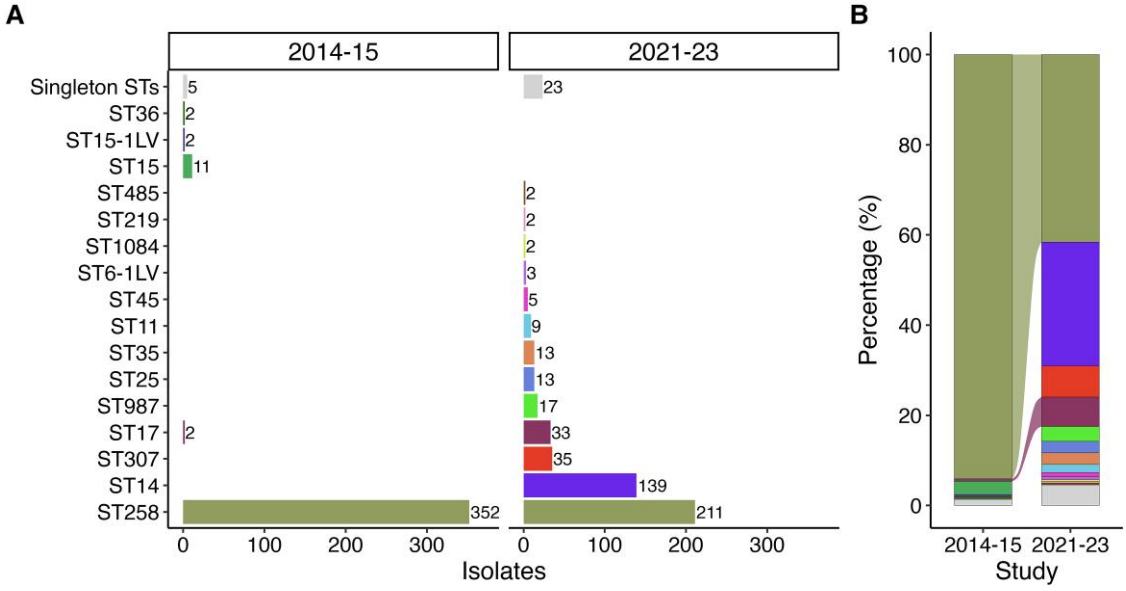


Figure 1. Increased diversity of carbapenem-resistant *Klebsiella pneumoniae* in California long-term acute care hospitals. *A*, Frequency of *K. pneumoniae* sequence types (STs) across study periods. *B*, Comparison of the distribution of STs across study periods. Sequence types that occurred only once were grouped as "Singleton STs" to aid visualization. Isolates with 1 allelic difference from a known ST, single-locus variants, were denoted with "–1LV." 1LV, single-locus variant; ST, sequence type.

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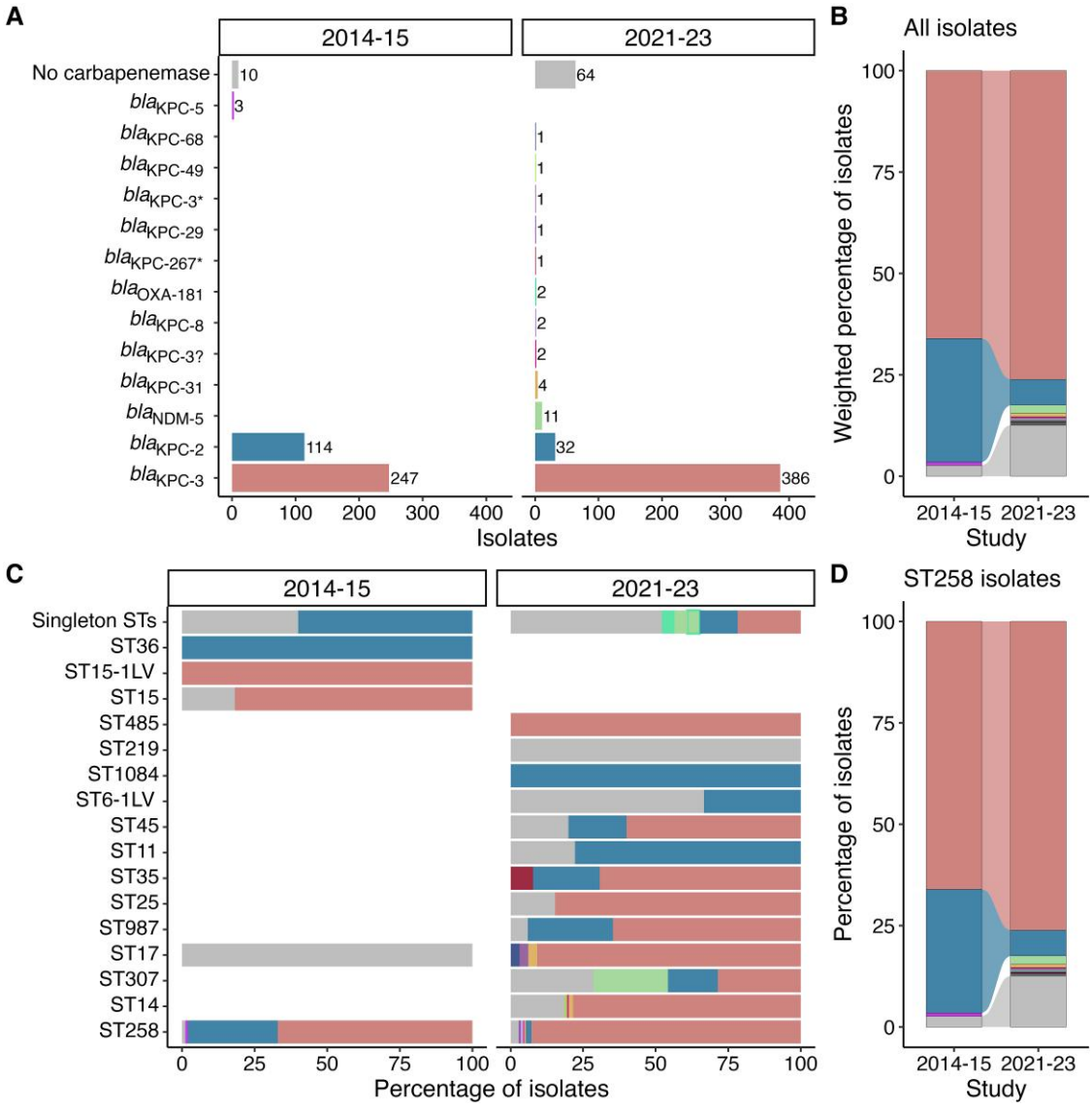


Figure 2. Diversification of carbapenemases across study periods. *A*, Frequency of carbapenemases among carbapenem-resistant *Klebsiella pneumoniae* isolates across study periods. One isolate harbored 2 carbapenemases. Genes with <100% identity were denoted with the "*" symbol, whereas genes with <100% coverage were denoted with the "?" symbol. *B*, Carbapenemase distribution across study periods. To simplify data visualization, the contribution of the isolate harboring $bla_{\text{NDM-5}}$ and $bla_{\text{OXA-181}}$ was split equally. *C*, Carbapenemase distribution across sequence types (STs), stratified by study period. Sequence types that occurred only once were grouped as "Singleton STs" to aid visualization. Isolates with 1 allelic difference from a known ST, single-locus variants, were denoted with "−1LV." For the isolate harboring multiple carbapenemases, a border color was introduced to highlight the second carbapenemase. *D*, Carbapenemase distribution among carbapenem-resistant *K. pneumoniae* ST258 isolates across study periods. *, identity <100%; ?, coverage <100%; 1LV, single-locus variant; ST, sequence type.

expansion of several preexisting ST258 and ST17 lineages from the 2014–2015 study period (Supplementary Figures 2–3).

Diversification of Carbapenem Resistance-Associated Genotypes

As the fitness cost of resistance genotypes can vary across genetic backgrounds [41], we hypothesized that changes in ST diversity would alter the prevalence of carbapenem resistance-associated genotypes.

Differences in the prevalence of carbapenemase families were detected across study periods (Figure 2*A* and 2*B*). Although the *K. pneumoniae* carbapenemase (bla_{KPC}) remained the dominant carbapenemase in study period 2, its prevalence decreased (364 [97.3%] vs 431 [85.0%]; χ^2 test $P < .001$). Congruent with this decrease was the detection of additional carbapenemase families at low frequencies: bla_{NDM} (11 [2.2%]) and bla_{OXA} (2 [0.4%]). The frequency of isolates without carbapenemases also increased in study period 2 (10 [2.7%] vs 64 [12.6%]; χ^2 $P < .001$). Changes in the distribution of bla_{KPC} alleles were also detected. Specifically, the prevalence of $bla_{\text{KPC-3}}$ (247 [66.0%] vs 386 [76.1%]) increased in study

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period 2, alongside a marked decrease in the prevalence of bla_{KPC-2} (114 [30.5%] vs 32 [6.3%]).

To ensure consistency of results across the broad, clinically defined CRKP population, we performed a sensitivity analysis using a more stringent definition of CRKP, focusing on isolates with SENSITITRE-confirmed resistance to imipenem and/or meropenem. Differences in ST and carbapenemase diversity persisted across study periods under this more restrictive definition of carbapenem resistance (Supplementary Figure 4).

The increase in noncarbapenemase-producing isolates led us to investigate the prevalence of alternative mechanisms of carbapenem resistance (Supplementary Figure 5). Although mutations in the major porins, *ompK35* and *ompK36*, were common across study periods, their frequency decreased in study period 2: *ompK35* (354 [94.7%] vs 393 [77.5%]; χ^2 $P < .001$) and *ompK36* (156 [41.7%] vs 110 [21.7%]; χ^2 $P < .001$). Porin mutations were common in isolates that did not harbor carbapenemases: *ompK35* (period 1, 5/10 [50.0%]; period 2, 43/64 [67.2%]) and *ompK36* (period 1, 4/10 [40.0%]; period 2, 20/64 [31.2%]). Virtually all noncarbapenemase-producing isolates had other β-lactamases (period 1, 10/10 [100%]; period 2, 62/64 [96.9%]). Most had at least one β-lactamase and a porin mutation (period 1, 7/10 [70.0%]; period 2, 52/64 [81.2%]). Across both study periods, the co-occurrence of porin mutations and β-lactamases often clustered by ST (Figure 2*C*; Supplementary Figures 6 and 7).

Given the abundance of epidemic lineage ST258 isolates in both study periods, we next compared the diversity of carbapenem resistance-associated genotypes (Figure 2*D*). Supporting the presence of both major circulating ST258 clades in the 2014–2015 study period [16, 20, 42], bla_{KPC-2}-containing (109 [31.0%]) and bla_{KPC-3}-containing (236 [67.0%]) isolates were identified. Conversely, virtually all 2021–2023 ST258 genomes harbored bla_{KPC-3} (196 [92.9%]), although a more diverse repertoire of minor bla_{KPC} alleles was detected. A statistically significant reduction in the presence of *ompK36* mutations was also observed among 2021–2023 ST258 genomes (153 [43.5%] vs 34 [16.1%]; χ^2 test $P < .001$). Inspection of the mutational profiles revealed the loss of an abundant bla_{KPC-2}-containing lineage with a glycine-aspartate (GD) insertion in the loop 3 of the Ompk36 porin (99 [28.1%] vs 4 [1.9%]; Fisher's exact test $P < .001$), coinciding with the near-total absence of major clade I of ST258 in the 2021–23 collection (Supplementary Figures 2, 3, 5, 6, and 7).

Dynamics of Carbapenemase-Containing Plasmids Reflect Changes to Population Structure

Next, we investigated whether changes in ST diversity reflected changes in the prevalence of circulating carbapenemase-containing plasmids. Among the 807 carbapenemase-containing isolates (period 1, 364; period 2, 443), 23 carbapenemase-containing plasmid clusters were identified. MOB-Suite plasmid clusters are characterized in Supplementary Figures 8 and 9.

Although 10 plasmids were identified in both study periods, the distribution of carbapenemase-containing plasmids was statistically different (Fisher's exact test $P < .001$) (Figure 3*A* and 3*B*). Several plasmids increased in prevalence in study period 2: AA356 (ColRNAI_rep_cluster_1857-like plasmid; 38 [10.4%] vs 94 [21.2%]) and AA038 (IncX3-like plasmid; 3 [0.8] vs 31 [7.0%]). Conversely, the prevalence of several resistance plasmids decreased in study period 2, notably AA018 (IncFII/IncFIB/rep_cluster_2183-like plasmid; 95 [26.1%] vs 3 [0.7%]), AA552 (IncN-like plasmid; 23 [6.3%] vs 9 [2.0%]), AA357 (IncFIA-like plasmid; 43 [11.8%] vs 3 [0.7%]), and AA275 (IncFIB/IncFII/rep_cluster_2183-like plasmid; 21 [5.8%] vs 10 [2.3%]). The prevalence of chromosomally encoded carbapenemase-producing isolates increased in study period 2 (16 [4.4%] vs 133 [30.0%]). Alongside differences in MOB-Suite plasmid clusters, differences in replicon type distributions were also observed across study periods (Supplementary Figure 10).

Changes in plasmid dynamics were tightly linked to changes in population structure across study periods (Supplementary Figures 9 and 11). First, the increased prevalence of chromosomal carbapenemases was driven by select STs, notably ST14 (Figure 3*C*). Second, the ST258 carbapenemase-producing population (period 1, 348; period 2, 205) experienced considerable changes to the diversity of its circulating plasmids. Although 9/12 (75%) plasmids were identified in both periods, their distributions differed (Figure 3*D*). Finally, considerable carbapenemase-containing plasmid sharing occurred across STs, with several abundant ST258 bla_{KPC-3}-associated plasmids, particularly AA356 and AD548 (rep_cluster_14818-like plasmid), being detected in minor STs (Figure 3*C*; Supplementary Figure 11).

Differences in Resistance Across Study Periods

To evaluate whether changes in ST diversity reflected a changing epidemiology of antibiotic resistance, we measured resistance to several important antibiotics using broth microdilution.

In the 2021–2023 study, virtually all isolates were resistant to the cephalosporinase-inhibiting BL/BLI combination, ceftolozane–tazobactam (486/507 [95.9%]), and the recently approved fluoroquinolone delafloxacin (481/507 [94.9%]). Resistance to recently approved tetracycline antibiotics was detected: eravacycline (112/507 [22.1%]) and omadacycline (100/507 [19.7%]). Resistance to cefiderocol was less common (29/505 [5.7%]). Susceptibility testing distributions, correlations among minimum inhibitory concentrations, and cross-resistance patterns are presented in Supplementary Figures 12–14, respectively.

Statistically significant differences were detected between study periods (Figure 4*A*). Although no statistical differences in resistance to KPC-inhibiting BL/BLI agents were detected across study periods (50/374 [13.4%] vs 68/507 [13.4%]; χ^2 $P = 1.00$), differences were observed for specific agents.

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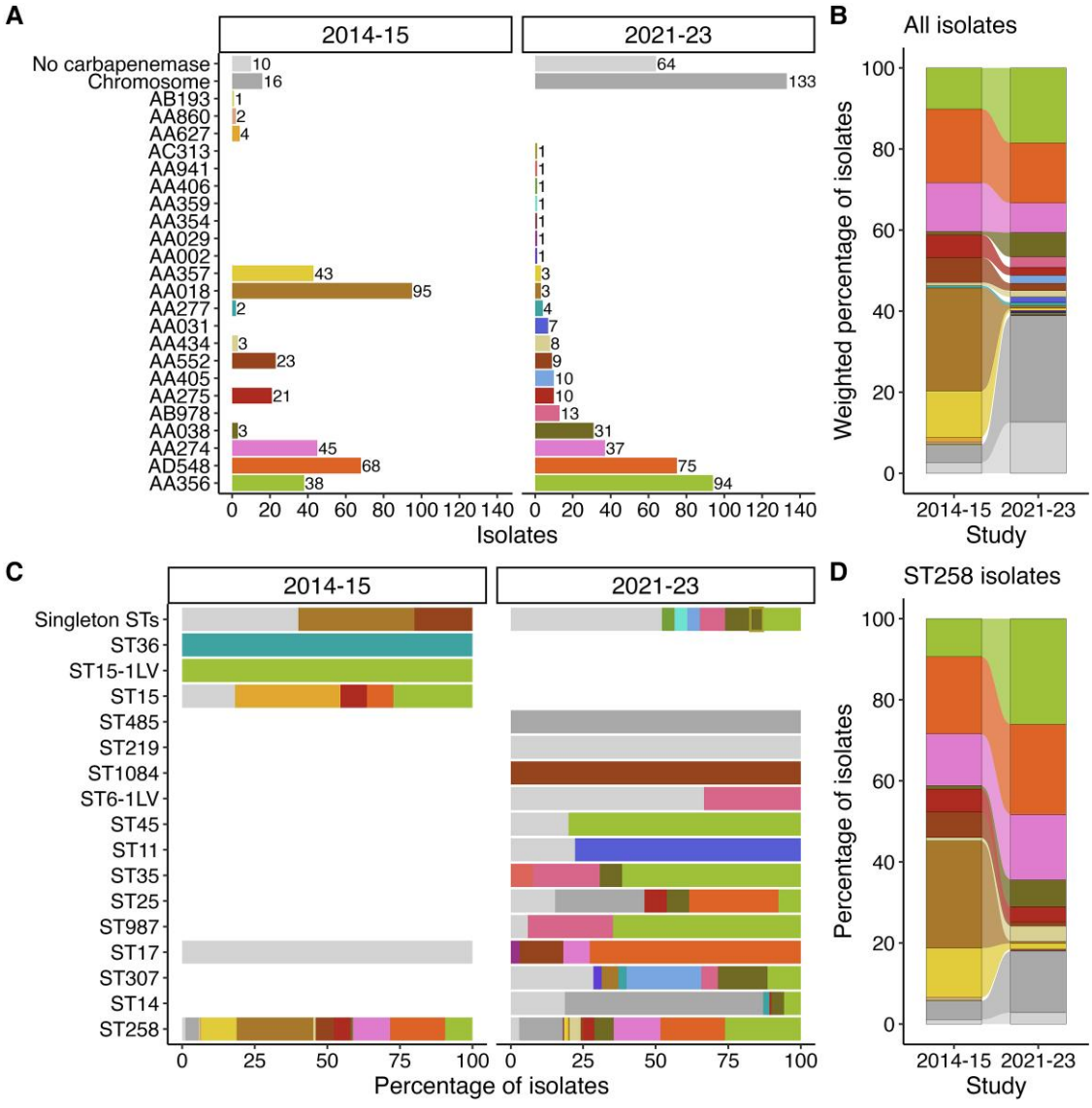


Figure 3. Changing abundance and diversity of carbapenemase-containing plasmids. *A*, Frequency of MOB-Suite-inferred carbapenemase-containing plasmids among carbapenem-resistant *Klebsiella pneumoniae* isolates across study periods. One isolate contained 2 carbapenemase-containing plasmids. *B*, Distribution of carbapenemase-containing plasmids across study periods. To simplify data visualization, the contribution of the isolate harboring 2 carbapenemase-containing plasmids was split equally. *C*, Distribution of carbapenemase-containing plasmids across sequence types (STs), stratified by study period. STs that occurred only once were grouped as "Singleton STs" to aid visualization. Isolates with 1 allelic difference from a known ST, single-locus variants, were denoted with "−1LV." *D*, Distribution of carbapenemase-containing plasmids among carbapenem-resistant *K. pneumoniae* ST258 isolates across study periods. 1LV, single-locus variant; ST, sequence type.

Resistance to meropenem-vaborbactam was lower in study period 2 (35/374 [9.4%] vs 10/507 [2.0%]; χ^2 $P < 0.001$), while resistance to ceftazidime-avibactam increased (4/374 [1.1%] vs 46/507 [9.1%], Fisher's exact test $P < .001$). Although resistance to plazomicin was rare in the 2014–2015 study, a moderate increase in resistance was detected in the contemporary collection (2/374 [0.5%] vs 29/507 [5.7%]; Fisher's exact test $P < .001$). Finally, decreases in resistance to colistin (131/374 [35.0%] vs 102/506 [20.2%]; χ^2 $P < .001$) and fosfomycin (101/374 [27.0%] to 68/507 [13.4%]; χ^2 $P < .001$) were also detected in study period 2.

Differences in Resistance Were Driven by Subpopulations of Carbapenem-Resistant *K. pneumoniae*

We hypothesized that the changes in antibiotic resistance correlated with differences in ST diversity. As ST258 dominated both study periods, we first compared the resistance landscape within this population (Figure 4*B*). Although no difference in resistance to KPC-inhibiting BL/BLI agents was detected across study periods (49/352 [13.9%] vs 24/211 [11.4%], χ^2 $P = .46$), differences in specific agents were detected. Resistance to meropenem-vaborbactam decreased in study period 2 (35/352 [9.9%] vs 2/211 [0.9%], Fisher's exact test $P < .001$),

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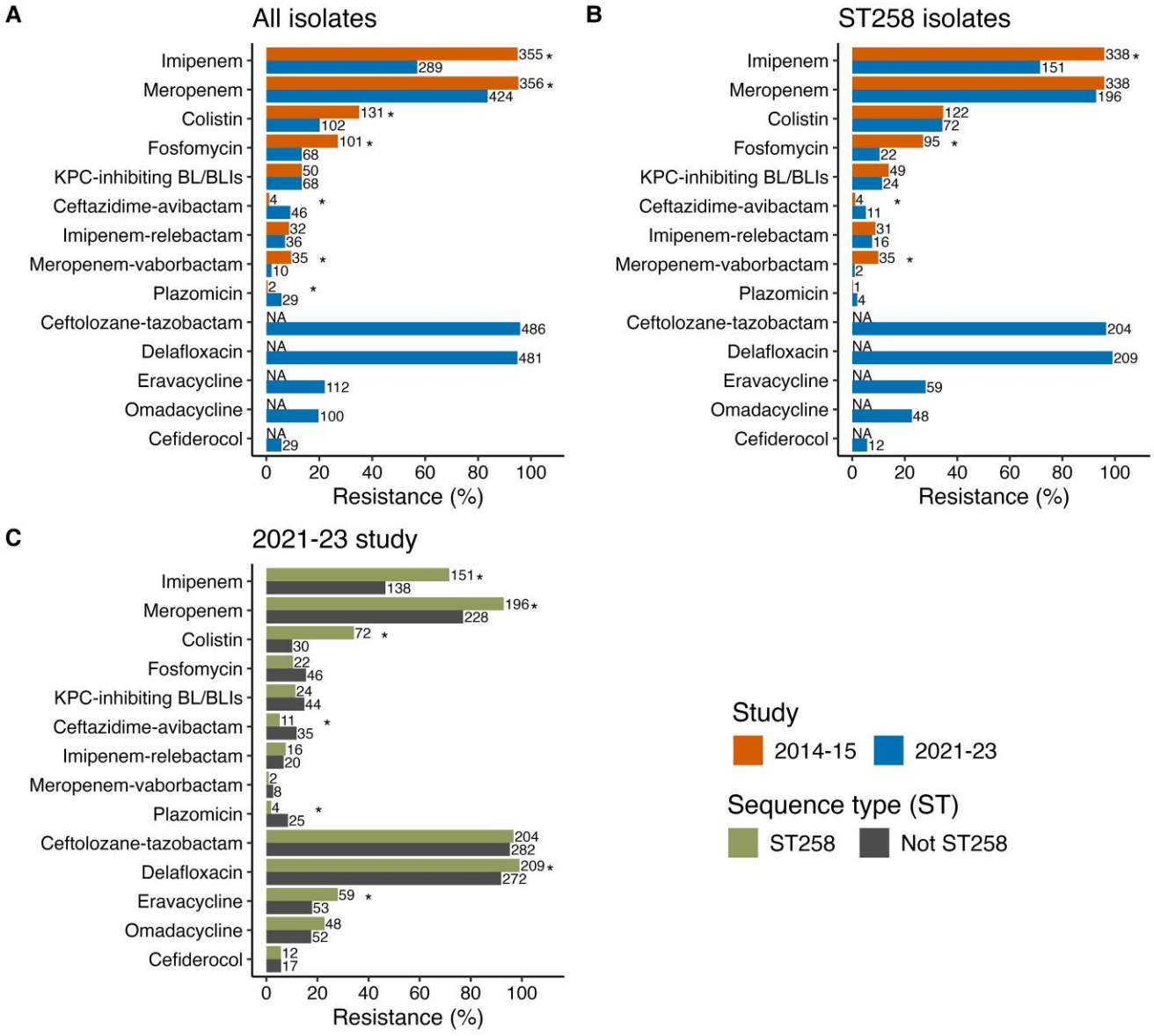


Figure 4. Differences in resistance to last-resort and recently approved antibiotics across study periods. *A*, Percentage of resistance to tested antibiotics across both study periods. *B*, Percentage of resistance to tested antibiotics among *Klebsiella pneumoniae* sequence type 258 (ST258) isolates obtained across study periods. *C*, Percentage of resistance to tested antibiotics among *K. pneumoniae* ST258 and non-ST258 isolates in the 2021–2023 study. The absolute number of resistant isolates is shown in the figure. In the 2021–2023 study, resistance to imipenem was defined as a minimum inhibitory concentration in the resistant category because the broth microdilution panel did not include concentrations with susceptible interpretations. Resistance to *K. pneumoniae* carbapenemase-inhibiting β-lactam/β-lactamase inhibitors is a composite outcome that indicates resistance to at least one of the following antibiotics: ceftazidime-avibactam, imipenem–relebactam, or meropenem–vaborbactam. *Statistically significant $P < .05$; BL/BLI, β-lactam/β-lactamase inhibitors; KPC, *K. pneumoniae* carbapenemase; NA, not tested in the 2014–2015 study; ST258, sequence type 258.

whileresistance to ceftazidime-avibactam increased (4/352 [1.1%] vs 11/211 [5.2%], Fisher's exact test $P = .006$). A statistically significant decrease in fosfomycin resistance was also observed in study period 2 (95/352 [27.0%] vs 22/211[10.4%], χ^2 $P < .001$). Contrary to the overall population-level decrease in colistin resistance across study periods, no statistically significant decrease was detected among ST258 isolates (122/352 [34.7%] vs 72/210 [34.3%], χ^2 $P = 1.00$).

These findings led us to hypothesize that the emergent, non-ST258 lineages contributed to the differences in resistance observed between the 2 study periods. To test this hypothesis, we compared the frequency of resistance to several novel antibiotics between ST258 and non-ST258 genomes collected from the 2021–2023 study (Figure 4C). ST258 isolates had elevated frequencies of resistance to delafloxacin (209/211 [99.1%] vs 272/296 [91.9%]; Fisher's exact test $P = .001$) and colistin (72/210 [34.3%] vs 30/296 [10.1%]; χ^2 test $P < .001$) than

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non-ST258 genomes. Conversely, non-ST258 genomes had a higher frequency of plazomicin resistance than ST258 genomes (4/211 [1.9%] vs 25/296 [8.4%]; Fisher's exact test $P = .002$).

Inspection of resistance patterns across STs in study period 2 revealed tight associations between resistance and select circulating lineages (Supplementary Figure 15). First, ST258 isolates accounted for most of those exhibiting resistance to colistin (72/102 [70.6%]), eravacycline (59/112 [52.7%]), and omadacycline (48/100 [48.0%]) and KPC-inhibiting BL/BLI combinations (24/68 [35.3%]). Second, ST17 accounted for a large proportion of isolates exhibiting resistance to cefiderocol (13/29 [44.8%]), fosfomycin (28/68 [41.2%]), and KPC-inhibiting BL/BLI combinations (18/68 [26.5%]). Third, the abundant epidemic lineage, ST14, had relatively low rates of resistance to the tested antibiotics. Finally, numerous minor STs had elevated resistance to select antibiotics, highlighting the potential for resistance to emerge across diverse genetic backgrounds. Phylogenetic investigation confirmed that resistance profiles were largely shaped by STs displaying differential propensities of resistance emergence and spread (Supplementary Figure 16).

DISCUSSION

Amidst the increasing burden of antibiotic resistance, it is critical to understand the evolving epidemiology of multidrug-resistant organisms. Our comparative analyses revealed an increased diversity of CRKP in our California LTACH network, driven by numerous regional introductions and clonal expansions of epidemic lineages. These changes coincided with different patterns of resistance genotypes and phenotypic resistance. Collectively, this work highlights the potential for repeated, comprehensive surveillance to inform regional public health initiatives by improving understanding of the shifting molecular epidemiology of multidrug-resistant organisms.

Considerable changes in the molecular epidemiology of CRKP were detected across study periods. Previously dominated by epidemic lineage ST258, ST diversity increased in the 2021–2023 study period. These changes reflected the introduction of other epidemic multidrug-resistant nosocomial lineages (ie, ST14 and ST307) and clonal expansion of the then-less-abundant lineage ST17. Similar to our findings, a recent study reports a shifting landscape of CRKP diversity among publicly available genomes, with the once-dominant, globally disseminated lineage ST258 being replaced by other STs [43]. We show that this trend also exists in LTACHs, a critical reservoir for CRKP and other multidrug-resistant organisms [3].

Inspection of ST258 genomes revealed considerable shifts in its population structure. Virtually all 2021–23 study isolates belonged to bla_{KPC-3}-containing clade II lineages, which maintained an elevated frequency of colistin resistance. Coinciding with these clonal expansions was the loss of a bla_{KPC-2}-harboring clade I lineage carrying a GD insertion in loop 3 of the OmpK36 porin. Prior investigation in this population revealed considerable clade-specific differences in anatomical site distribution and the propensity for resistance evolution and spread [16, 17, 44]. Longitudinal studies have also supported the dynamic nature of ST258, with adaptive mutations, changes in accessory genome content, the introduction of novel lineages, and the stability of resistance plasmids influencing the dynamics of circulating lineages [12, 45, 46]. Future work should determine the contribution of strain replacement, clonal expansion, and resistance evolution in shaping the evolutionary trajectory of ST258 in this region. Additionally, future work should investigate how facility-wide antibiotic consumption patterns, infection prevention practices, and changes in patient population characteristics influence the epidemiology and evolution of CRKP lineages.

The shifting characteristics of CRKP represent a pivotal nexus in understanding and predicting resistance. Despite their introduction over the past 10 years, our data demonstrate reasonable retention of susceptibility for recent novel antibiotics (ie, KPC-inhibiting BL/BLI agents, recently approved tetracyclines, and cefiderocol). This indicates that, at least to date, the introduction of these drugs has not led to appreciable emergence of resistance to these agents. Our study also identified reductions in resistance to several older antibiotics (ie, fosfomycin, colistin, and plazomicin), presumably due to reduced selective pressure from antibiotic stewardship programming and decreased usage following the introduction of newer agents. These hypotheses should be tested in larger studies examining the correlation between facility-level antibiotic consumption and resistance patterns.

Inspection of resistance patterns across STs revealed the potential for distinct lineages to drive the emergence and spread of resistance to clinically important antibiotics. Early recognition of epidemic STs with elevated potential for resistance development and subsequent cross-transmission could aid in forecasting regional- or facility-level trends and directing appropriate infection prevention interventions to curb regional transmission. Integration of intensive, regional molecular surveillance with commonly used antibiotic stewardship strategies, such as regional antibiograms and monitoring antibiotic prescribing patterns [47], will enable clinicians and public health practitioners to design tailored interventions to slow the development of resistance.

Our study has several limitations. First, longitudinal surveillance was not performed during 2015–2021, the period during which several new antibiotics were introduced. Second, restriction to the patient's first clinical isolate per quarter minimized the contribution of subsequent infection episodes to our analyses (Supplementary Figure 1). This was chosen to minimize bias from differences in culturing rates. Third, we only evaluated changes in clinical isolates. As gut colonization often precedes infection [48], colonized patients can be an integral contributor to the regional burden of CRKP.

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In conclusion, molecular surveillance across 2 time periods illuminated the changing epidemiology of CRKP in California LTACHs. Our regional surveillance study highlights how shifts in CRKP diversity shape circulating resistance genotypes and drive the emergence and spread of resistance. Future studies should investigate the influence of changing antibiotic formularies on the evolution and epidemiology of multidrug-resistant organisms across diverse geographic and healthcare contexts.

Supplementary Data

Supplementary materials are available at *The Journal of Infectious Diseases* online (http://jid.oxfordjournals.org/). Supplementary materials consist of data provided by the author that are published to benefit the reader. The posted materials are not copyedited. The contents of all supplementary data are the sole responsibility of the authors. Questions or messages regarding errors should be addressed to the author.

Notes

Author contributions. K. J. G., E. S. S., B. J. K., L. G., J. H. H., E. J. C. G., and E.L. conceived and designed the study. A. J.- B., L. C., L. L. C., and A.P. facilitated the collection and processing of isolates and clinical data. E. S. S. and E. L. supervised the research. L.C. and W. B. B. performed clinical analyses. K. J. G. performed computational analyses. K. J. G., E. S. S., and E. L. wrote the manuscript with input from all coauthors. All authors read and approved the final manuscript.

Ethical approval. This study was reviewed and approved by the Institutional Review Board of the University of Pennsylvania.

Data availability. The datasets used and/or analyzed are available from the corresponding author on reasonable request.

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Potential conflicts of interest. J. H. H. was affiliated with the University of Pennsylvania during the conduct of this research and is currently an employee of AstraZeneca. The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the CDC or the National Institutes of Health. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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